

Yuhao (David) Cao

Greater Toronto Area, Ontario, Canada · davidyh.cao@mail.utoronto.ca · (289) 885-9268
[linkedin.com/in/yuhao-cao](https://www.linkedin.com/in/yuhao-cao) · github.com/davidcaoyh · yuhaocao.com

EXPERIENCE

Master of Engineering, ECE, University of Toronto 2025 – 2028 (expected)

Coursework: ECE 1508 Deep Generative Models · ECE 1513 Introduction to Machine Learning · APS 1070 Foundations of Data Analytics and Machine Learning · ECE 1779 Introduction to Cloud Computing · ECE 1778 Creative Applications for Mobile Devices

Consultant at Deloitte 2021 – Current

Adobe Digital Marketing expert in Content Management and Customer Data Platform solutions

Bachelor of Science, Computer Science, McGill University 2017 – 2021

Hugh Brock Scholarship, awarded for academic excellence and leadership.

ACADEMIC COURSE PROJECTS

AI Village - a continuously running multi-agent LLM System Summer 2026

<https://ai-village.yuhaocao.com/> · <https://github.com/davidcaoyh/ai-village>

- Built the agent runtime in Python: a function-calling loop over OpenRouter driving four models from four providers, with a nine-tool dispatcher spanning web search, HTTP fetch, DOI resolution, sandboxed file I/O, inter-agent chat and turn control.
- Built the front end website using a FastAPI service over a SQLite event log recording every model call, tool call and token cost; Deployed on DigitalOcean Droplet
- Customized persona for each agent; implemented a unanimous commitment-vote termination protocol.

Does Multimodal Retrieval Reduce Hallucination in News Summarization? Summer 2026

A Text-vs-Pixel RAG Study · ECE 1508, Deep Generative Models, University of Toronto

github.com/davidcaoyh/multimodal-news-rag

- Built a dual-encoder multimodal RAG pipeline over 1,023 BBC News article–image pairs: SBERT for passages, CLIP for images, both in FAISS, with the two scores combined at a weight tuned on a held-out split.
- Wrote the evaluation harness: an LLM judge that breaks summaries into atomic claims and checks text and image support separately, scored with paired bootstrap tests.
- Designed an ablation study to isolate whether multimodality reduces hallucination, rather than retrieval in general; found the gain comes from retrieval alone—claims supported by retrieved evidence rose from 15% to 86% (+0.59 paired), while both pre-registered multimodal steps were null (+0.019 and +0.025, $p = 0.64$ and 0.10).

Short-Term Urban Traffic Speed Prediction Winter 2026

ECE 1513, Introduction to Machine Learning · group project, NeurIPS-format report

<https://github.com/davidcaoyh/ECE1513ML>

- Built the dataset and feature pipeline: merged 2.1M Toronto road-sensor records over 1,947 segments with hourly Environment Canada weather into 1.58M time-aligned rows with 45 features
- Benchmarked six ML models against naive baselines for 15-minute traffic speed prediction; demonstrated feature engineering cut MAE 16–23% across tree, distance, and neural families while tuning added under 0.5%; delivered MAE 3.180 km/h and R^2 0.781 with XGBoost — 24.4% over the Persistence baseline, on par with a ResNet MLP at 3× faster training.

TECHNICAL SKILLS

Python, Java, JavaScript, SQL, PyTorch, CI/CD Pipeline, GitHub